Milestone 2 – Data Visualization (COM-480)

Through the Lens: 4 Years of Campus Life in 57k Photos

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Our dataset consists of 56,890 photographs taken over four years on the EPFL campus, accompanied by 952 metadata fields: capture parameters (ISO, aperture, shutter speed, focal length), temporal context, camera and lens identifiers, and Lightroom post-processing adjustments. Rather than analyzing the images themselves, we turn this invisible technical layer into a visual narrative about campus life and creative evolution.

The website follows a scroll-driven structure, guiding the user from broad temporal patterns toward increasingly technical photographic detail. It is live at `https://com-480-data-visualization.github.io/through_the_lens/`.

Visualizations

Visualization 1 – Photo Activity

Shooting volume is not uniform across time: some years and days concentrate far more photographs than others. This visualization asks: *when does shooting happen, and does that rhythm repeat across years?* The design is a single interactive panel with two drill-down levels. At the top level, a vertical bar chart shows annual photo counts (Figure 1); clicking a bar transitions to a GitHub-style calendar heatmap for that year (Figure 2).

Year view. Each bar represents one year, coloured with a distinct series colour and highlighted in amber at the peak. Partial years (2021 and 2026) are rendered at reduced opacity to avoid misleading comparisons.

Day view. Columns are calendar weeks, rows are days of the week (Monday–Sunday), and cell colour encodes daily photo count on a sequential blue gradient. Clicking a cell opens a side panel showing the date, photo count, and the name of the shooting session that day.

Core MVP: Two-level drill-down (year → day) with animated transitions.
Extra ideas: A thumbnail of a representative photograph from that day displayed in the side panel, connecting the abstract count to the actual visual content.
Tools: Currently implemented with Canvas 2D API; planned migration to D3.js for richer axis transitions and smoother animated drill-down between the year view and the day heatmap.

Courses: Data (4.1) Interactions (5.1)

Visualization 2 – Gear Timeline

Camera equipment changed significantly over the four years, spanning compact cameras, phones, and several mirrorless bodies. This visualization asks: *which camera body was active during each period, and how abruptly did transitions happen?*

The chart is a swimlane heatmap: one horizontal row per camera body, one column per month (September 2021 – December 2024), with cell colour intensity proportional to the number of shots taken that month by each body (normalised per body so that all cameras remain visible regardless of total volume). Empty cells are rendered as a faint track, making gaps in usage immediately apparent. The user can

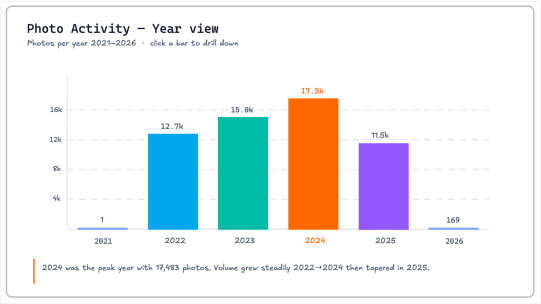

Figure 1: Year view: annual photo counts 2021–2026. Peak year (2024, 17,483 photos) highlighted in amber; partial years at reduced opacity.

Figure 2: Day view: full-year calendar heatmap for 2024. Blue intensity encodes daily photo count; clicking a cell surfaces the date, count, and session name.

toggle individual bodies on or off via labelled buttons, and hovering over any cell reveals the exact shot count and month.

This static design was chosen over an animated bar-chart race because the data is strongly skewed toward a single body (Sony A7 IV, ≈96% of shots); a race would collapse to one dominant bar. The swimlane preserves the full temporal picture: the Sony dominates throughout, the Panasonic LX100 appears briefly in early 2023, the DJI drone in summer 2023, and the Fuji X100VI enters strongly from July 2024 onwards.

Core MVP: Static swimlane heatmap with per-body toggle and hover tooltips (shot count + month label).
Extra ideas: Click a cell to surface sample shots from that camera/month; colour-code cells by shooting location.

Courses: Perception & Colors (6.1), Tabular Data (11.1)

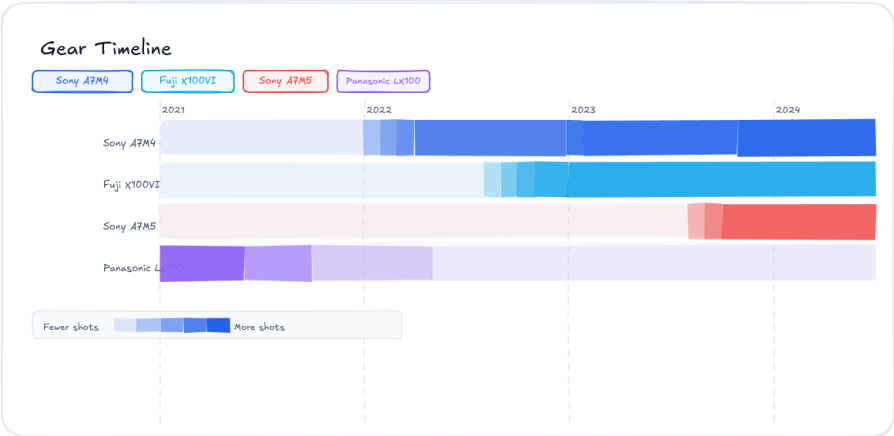

Figure 3: Gear Timeline: swimlane heatmap of monthly shot counts per camera body. Cell intensity is normalised per row so all bodies remain visible. Hover reveals exact counts and month.

Visualization 3 – Exposure Explorer

ISO, aperture, and shutter speed form the *exposure triangle*: adjusting one forces tradeoffs on the others. This visualization places all ≈57,000 photos as individual points in a two-axis space, letting the user choose which parameters to compare and discover correlations themselves.

The user picks X and Y axes from four parameters (ISO, aperture, shutter speed, focal length); when axes change, every point animates smoothly to its new position using spring physics. Points are colored by a third dimension (year, camera body, or scene type inferred from ISO level), and hovering reveals

full metadata. A contextual insight card explains in plain language what the selected pair of axes reveals – for instance, that plotting ISO against aperture exposes the classic dark-scene tradeoff, where both values are forced high simultaneously.

Core MVP: Scatter plot with selectable X/Y axes, color dimension, hover tooltips, and spring-animated axis transitions.
Extra ideas: Lasso selection to isolate a cluster; linked highlighting with the Gear Timeline.

Courses: Interactions (5.1), Tabular Data (11.1)

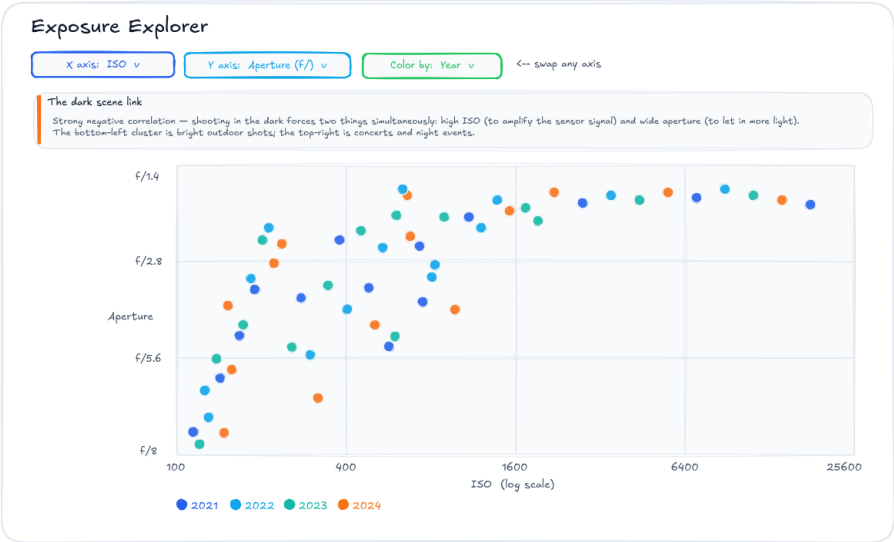

Figure 4: Exposure Explorer: each dot is one photo in a user-configured two-axis space. Switching axes animates every point to its new position via spring physics. A contextual insight card explains the selected combination.

Visualization 4 – Lens Race

This visualization asks: *how did the mix of lenses shift over time, and which focal lengths gained or lost favour?*

The top five lenses by total shot count are retained: the 70–200 mm f/2.8 GM II (telephoto zoom), 24–70 mm f/2.8 GM II (standard zoom), 50 mm f/1.4 GM (portrait prime), 14 mm f/1.8 GM (ultra-wide prime), and the 28–70 mm f/3.5–5.6 OSS (kit zoom).

The chart is an animated horizontal bar chart race (Figure 5): each bar represents a lens, its length encodes the cumulative shot count from the start of the timeline to the current month, and bars reorder dynamically as rankings change. A play/pause button and a scrub slider let the user control playback speed or jump directly to any month between February 2022 and December 2024.

Core MVP: Animated bar chart race with play/pause, scrub slider, and a colour legend.
Extra ideas: Overlay key events (new lens purchase, trip) as annotations on the timeline; filter by focal-length category (wide / standard / tele).

Courses: Interactions (5.1), Perception & Colors (6.1)

Visualization 5 – Workflow Sankey

As a complementary view, we added a compact Sankey diagram summarizing the workflow from *Picture shots* to *Selection*, *Edition*, and *Published*. It highlights both the kept flow and the drop-off at each step, with link widths proportional to counts and hover tooltips for exact values and rates.

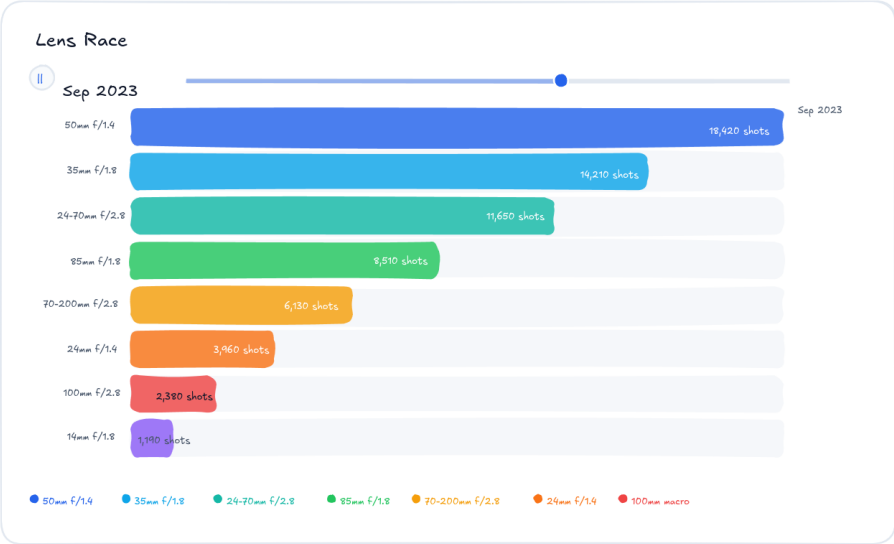

Figure 5: Lens Race: animated bar chart race of cumulative shots per interchangeable lens. Bars reorder as rankings change. Play/pause and scrub slider control playback.

Figure 6: Workflow Sankey: flow from *Picture shots* to *Selection*, *Edition*, and *Published*, with losses shown at each step. Link widths are proportional to counts; hover tooltips provide exact values and drop-off/conversion rates.

Website and Extra Ideas

- **Website name and identity.** "Through the Lens" reflects the idea of reading a photographer's evolution through the metadata hidden inside every image.
- **Color palette.** Light off-white background (`#F8FAFC`), white cards, orange accent (`#F97316`) for highlights, and a sequential blue palette (`#2563EB` → `#8B5CF6`) for data series. Typography: Playfair Display (headings, italic) and DM Sans (body, weight 300–500).
- **Navigation style.** Full-screen scroll-snap layout, a fixed navigation bar and right-side dot indicators that allow direct section jumping.

Additional ideas:

- **Lightroom Edit Fingerprint.** A radar chart showing the average Lightroom adjustment profile (exposure, contrast, clarity, dehaze, shadows) per lens or camera body.